



Australian
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Bucket Trigram

Extend the bucket bigram model to consider *two* words of context instead of one, leading to better text generation.

You will need

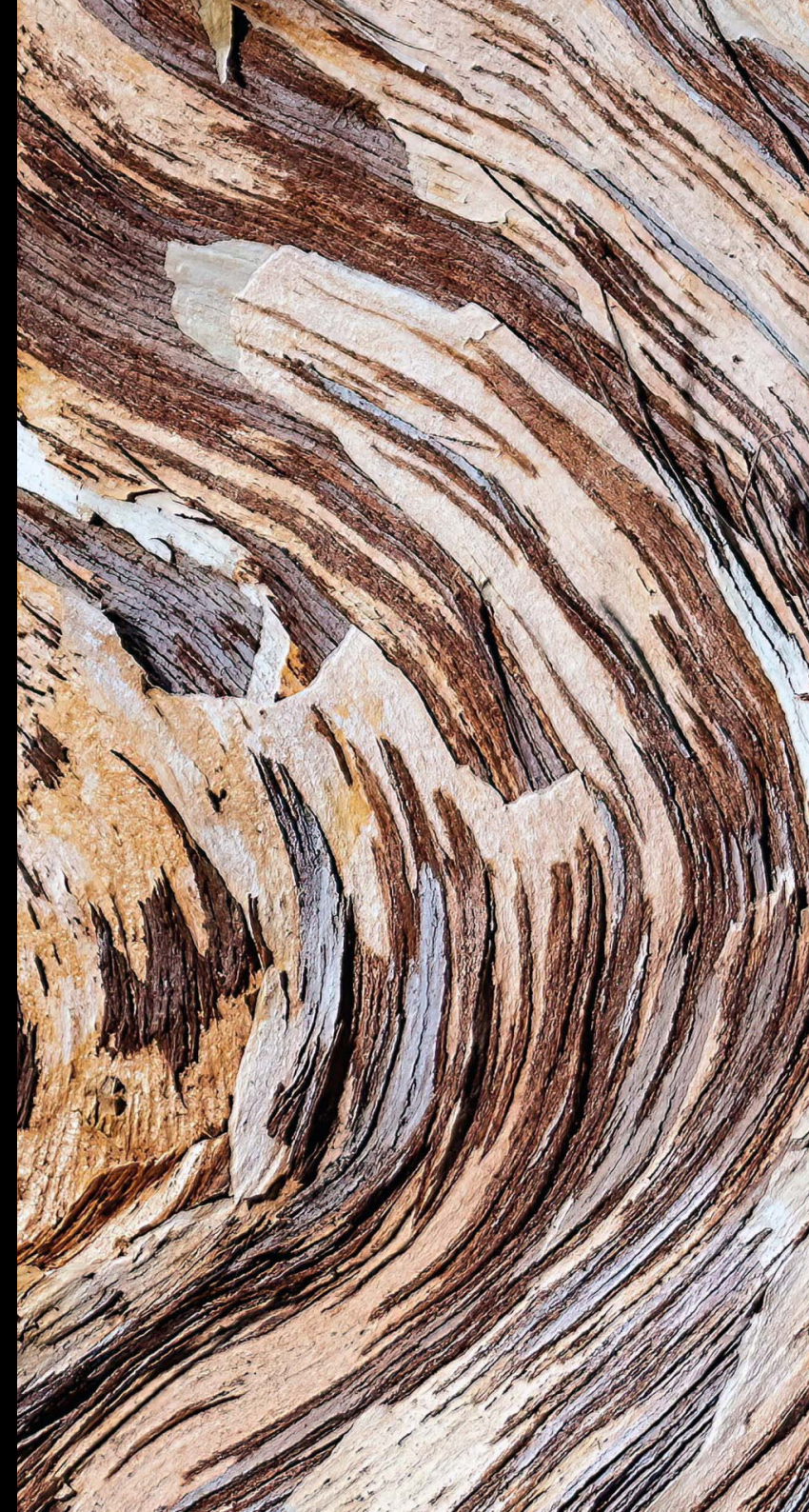
- the same materials as *Bucket Training*
- additional small containers for two-word label buckets
- sticky notes or paper for bucket labels (you'll need to write two words on each label)

Your goal

To build a trigram language model using buckets where each bucket is labelled with *two* words instead of one. **Stretch goal:** train on more data or generate longer outputs.

Key idea

Trigrams show how more context improves prediction quality. Instead of asking “what follows this word?”, we ask “what follows these *two* words?”. This means more buckets to manage, but better predictions.



Algorithm (training)

1. **prepare your tokens** as per *Bucket Training* (print, cut into tokens, keep in order)
2. **build the model** using word *pairs* as bucket labels:
 - take the first *two* tokens — these form your bucket label
 - create the bucket if needed, then put the *third* token inside it
 - shift along by one (new pair = old second word + token just placed)
 - repeat until all tokens are in buckets

Example (training)

Original text: “See Spot run. See Spot jump.”

Prepared tokens: see spot run . see spot jump .

After processing all tokens:

Bucket label	Tokens inside
see spot	run jump
spot run	.
run .	see
. see	spot
spot jump	.

The “see spot” bucket has two tokens because different words followed that pair. Compare to bigram where “see” would just contain spot spot.

Algorithm (generation)

1. **choose a starting bucket**; write down its two-word label
2. **close your eyes and pick a random token** from that bucket
3. **write down the token**, then put it back in the bucket
4. **find the bucket** whose label matches your last *two* words (second word of old label + the token you just picked)
5. if no bucket exists, use any bucket starting with the *first* word instead
6. **repeat** from step 2 until you reach a stopping point

Example (generation)

Using the bucket model from training:

1. start with see spot bucket; write “see spot”
2. pick randomly: run or jump — say we get run; write it
3. find bucket “spot run”; pick .; write it
4. find bucket “run .”; pick see; write it
5. find bucket “. see”; pick spot; write it
6. find bucket “see spot”; this time pick jump; write it
7. find bucket “spot jump”; pick .; write it

Generated text: “see spot run. see spot jump.”